

Power BI Mini Project

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Project Title: Sales Data Analysis Using Power BI

Dear Students,

Welcome to your sales data analysis project using Power BI! This project is designed to provide you with hands-on experience in analyzing business data and creating meaningful visual reports that can help guide decision-making. You will work with a real-world dataset that contains detailed sales records, including customer information, product categories, shipping methods, and sales performance metrics.

The primary goal of this project is to empower you with the skills to explore and visualize large datasets, uncover trends, and draw insights that can be used to make data-driven decisions. By the end of this project, you should have developed a deeper understanding of sales performance, customer behaviour, and market dynamics—all essential skills in the modern business landscape.

Learning Outcomes:

By completing this project, you will:

- Gain proficiency in using Power BI to manipulate and visualize data.
- Learn how to clean, transform, and enhance raw data in Power BI for analysis.
- Develop the ability to analyze sales trends, customer patterns, and product performance.
- Create dynamic, interactive dashboards that can tell a story through data.
- Understand how to ask the right business questions and find answers using data-driven insights.

Project Scope:

The dataset you are working with contains sales records from various customers across different regions and includes detailed information on sales, customer segments, shipping methods, and product categories. You will be expected to:

1. **Explore the Data:** Understand the structure and contents of the dataset. Identify key variables that will help you analyze sales and customer behavior.
2. **Create Visualizations:** Use Power BI's visual capabilities to build dashboards that showcase trends, patterns, and insights from the data.
3. **Analyze the Data:** Answer important business questions related to sales performance, customer segmentation, product analysis, and shipping methods.
4. **Provide Recommendations:** Based on your analysis, offer data-driven insights that could help improve sales strategies or operational efficiency.

Dataset: The dataset is available with the file name ***Dataset – Mini Project*** at the following URL:

<enter the URL here>

Data Overview:

The dataset provided is a sales dataset, containing detailed records of customer transactions. It has the following columns:

- **Row ID:** A unique identifier for each row.
- **Order ID:** A unique identifier for each order.
- **Order Date:** The date when the order was placed.
- **Ship Date:** The date when the order was shipped.
- **Ship Mode:** The method by which the product was shipped (e.g., Second Class, Standard Class).

- **Customer ID:** A unique identifier for each customer.
- **Customer Name:** The name of the customer.
- **Segment:** The customer segment (e.g., Consumer, Corporate).
- **Country:** The country where the order was placed.
- **City:** The city of the customer.
- **State:** The state of the customer.
- **Postal Code:** The postal code of the customer.
- **Region:** The geographical region (e.g., South, West).
- **Product ID:** A unique identifier for each product.
- **Category:** The product category (e.g., Furniture, Office Supplies).
- **Sub-Category:** The sub-category of the product (e.g., Bookcases, Chairs).
- **Product Name:** The name of the product.
- **Sales:** The total sales amount for that transaction.

Key Areas for Analysis:

This dataset provides rich information to explore and visualize sales performance, customer behavior, shipping patterns, and product analysis. Students can create various Power BI reports to answer important questions.

Potential Questions for Students to Explore:

1. **Sales Performance:**
 - What are the total sales over time (e.g., by month, quarter, or year)?
 - Which products, categories, or sub-categories generate the most revenue?
 - How do sales vary by region, state, or city?
2. **Customer Analysis:**
 - Which customer segments (Consumer, Corporate) generate the highest sales?
 - Who are the top customers by total sales?
 - What are the geographical trends in customer purchases?
3. **Shipping and Delivery Analysis:**
 - What is the most popular shipping mode?
 - What is the average delivery time (difference between order date and ship date)?
 - Are there any correlations between shipping mode and sales performance?
4. **Product Analysis:**
 - Which product categories and sub-categories are most and least profitable?
 - What are the top-selling products?
 - How does the performance of products vary by region or segment?
5. **Time-based Trends:**
 - Are there any seasonal patterns in sales?
 - How have sales grown or declined over the years?
6. **Regional Performance:**
 - Which regions or states have the highest and lowest sales?
 - How do different regions perform across different product categories?

Recommended Visualizations:

- **Bar charts** for comparing sales by product categories, sub-categories, and regions.
- **Line charts** to analyze sales trends over time.
- **Maps** to visualize sales distribution across states and cities.
- **Pie charts or Donut charts** for customer segments and shipping mode breakdowns.

- **Tables** to list top customers or products by sales.

Words of Caution:

While the recommended visualizations offer a helpful starting point, they are not meant to limit your creativity or dictate how you should approach the project. These suggestions aim to help you gain initial insights from the data, but the true value of analysis comes from pushing beyond the basics. Power BI offers a wide array of tools, so don't hesitate to experiment with advanced charts, slicers, calculated measures, and other features to add depth to your analysis.

This is your chance to showcase your creativity and critical thinking. Explore different questions, uncover hidden patterns, and find innovative ways to present your findings. Let your insights shape your visualizations, and remember that the goal is not only to meet expectations but also to push the boundaries of what's possible.

Final Thoughts:

This project is not just about building a dashboard—it's about thinking like a data analyst and using data to make informed business decisions. Whether your future involves data analysis, business intelligence, or any field that works with data, the skills you gain here will serve as a strong foundation.

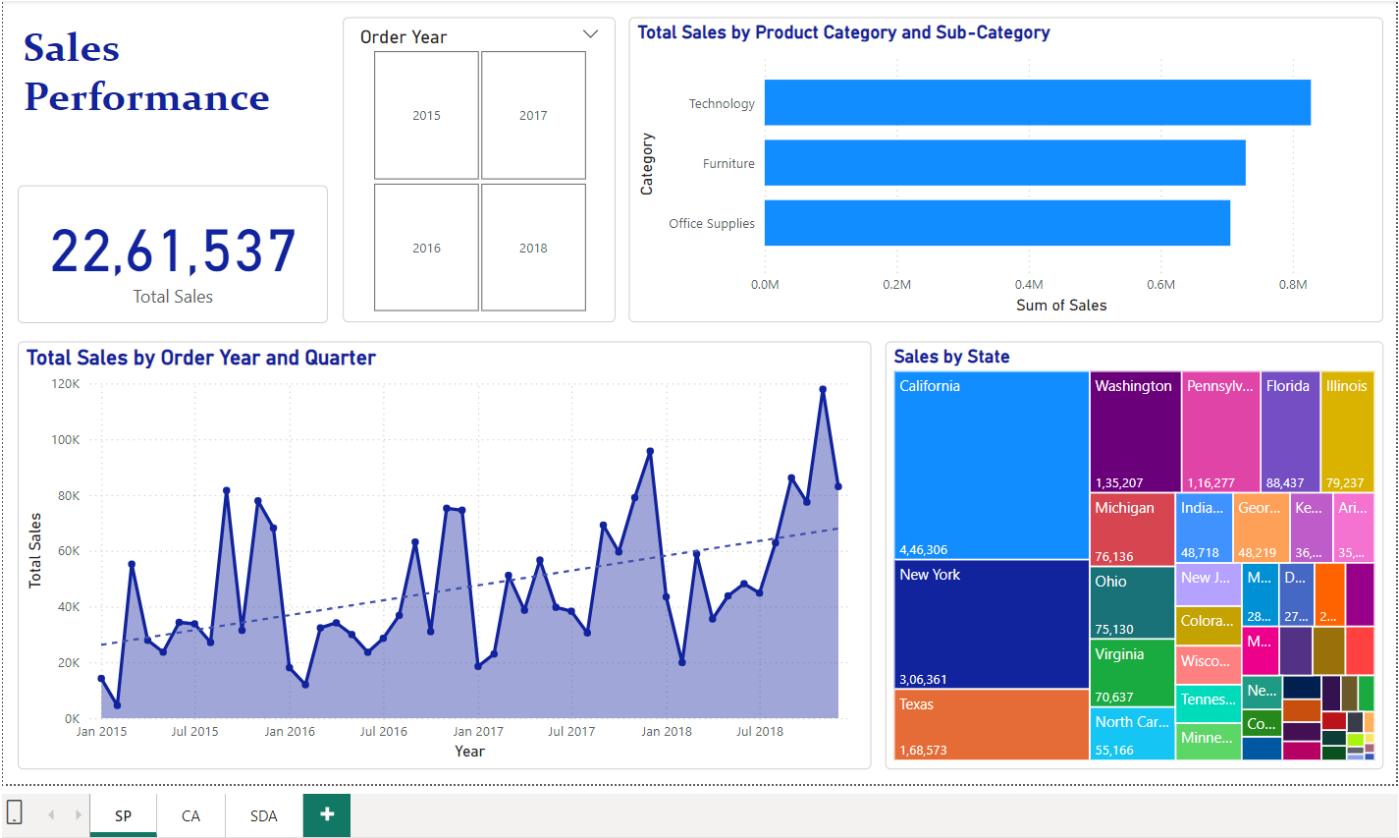
I look forward to seeing the creativity, analysis, and insights you uncover through this project. Have fun exploring the data, and good luck!

Best regards,

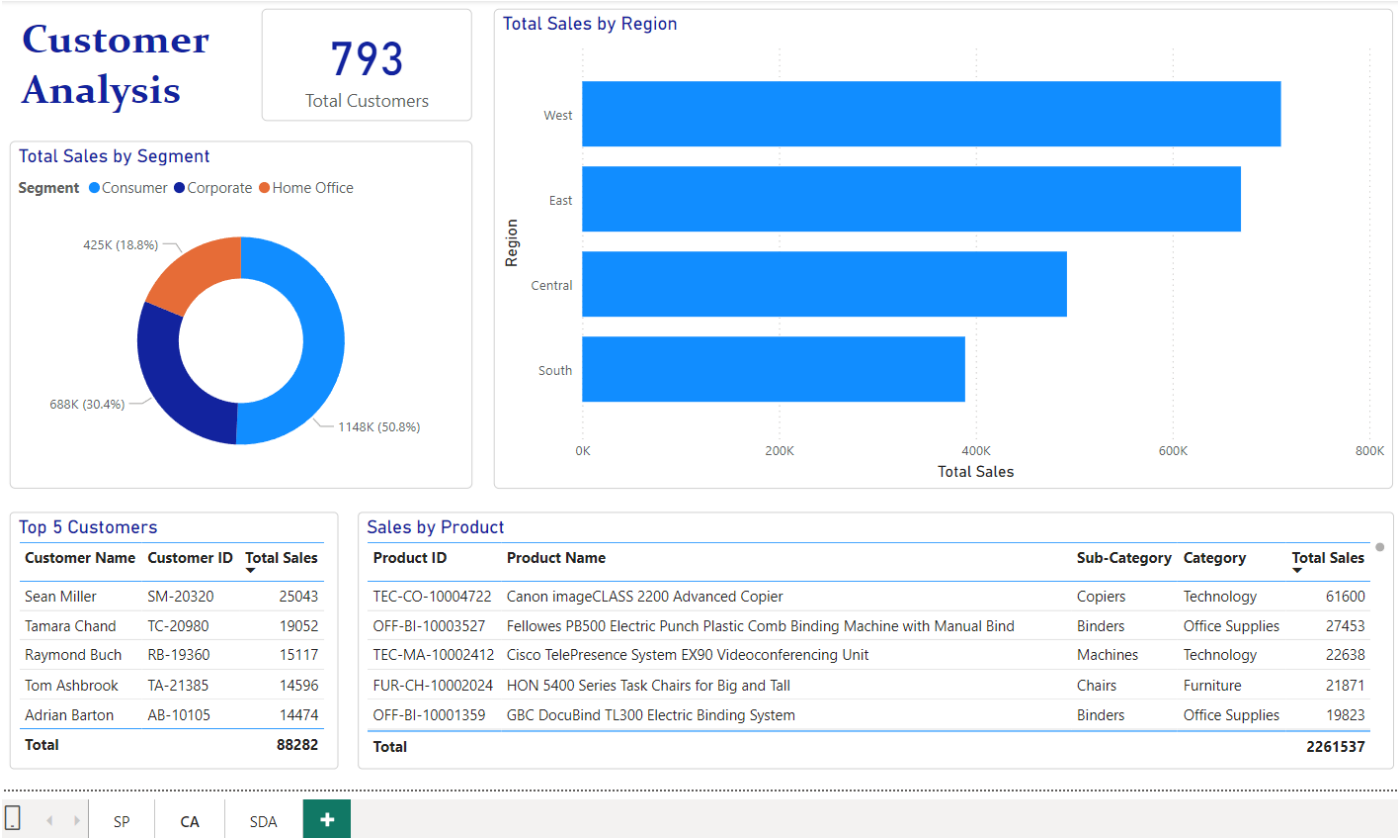
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Sample Dashboard screenshot has been given in the following pages for your reference.

Dashboard 1 Sample



Dashboard 2 Sample



Dashboard 3 Sample

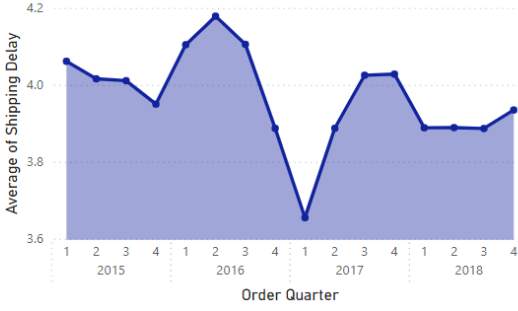
Shipping & Delivery Analysis

3.96
Average Shipping Delay (Days)

Sales by Ship Mode and Product Category

Ship Mode	Furniture	Office Supplies	Technology	Total
Standard Class	4,28,137.8	4,22,369.86	4,90,324.28	13,40,831.31
Second Class	1,52,657.84	1,56,054.42	1,41,201.92	4,49,914.18
First Class	1,08,739.05	98,227.44	1,38,605.77	3,45,572.26
Same Day	39,124.51	28,770.63	57,323.91	1,25,219.04
Total	7,28,658.58	7,05,422.33	8,27,455.87	22,61,536.78

Average of Shipping Delay by Order Year and Order Quarter



Order Count by Shipping Delay in Days



Shipping Mode Distribution by Order Count

